cut. All components of the stabilizers are purchased from outside sources , assembly and testing is carried out at company's own works. The assembly and testing time required for the two models are 0.8 hour each for ordinary and 1.20 hours each for auto-cut. Manufacturing capacity 720 hours at present is available per week. The market for the two models has been surveyed which suggests maximum weekly sale of 600 units of ordinary and 400 units of auto-cut. Profit per unit for ordinary and auto-cut models has been estimated at ₹ 100 and ₹ 150 respectively. Formulate the linear programming problem.

4. Solve the following linear programming problems by graphical method.
   (i) Maximize $Z = 6x_1 + 8x_2$ subject to constraints $30x_1+20x_2 \leq 300$; $5x_1+10x_2 \leq 110$; and $x_1, x_2 \geq 0$.
   (ii) Maximize $Z = 22x_1 + 18x_2$ subject to constraints $960x_1 + 640x_2 \leq 15360$; $x_1 + x_2 \leq 20$ and $x_1, x_2 \geq 0$.
   (iii) Minimize $Z = 3x_1 + 2x_2$ subject to the constraints $5x_1 + x_2 \geq 10$; $x_1 + x_2 \geq 6$; $x_1 + 4x_2 \geq 12$ and $x_1, x_2 \geq 0$.
   (iv) Maximize $Z = 40x_1 + 50x_2$ subject to constraints $3x_1 + x_2 \leq 9$; $x_1 + 2x_2 \leq 8$ and $x_1, x_2 \geq 0$
   (v) Maximize $Z = 20x_1 + 30x_2$ subject to constraints $3x_1 + 3x_2 \leq 36$; $5x_1 + 2x_2 \leq 50$; $2x_1 + 6x_2 \leq 60$ and $x_1, x_2 \geq 0$
   (vi) Minimize $Z = 20x_1 + 40x_2$ subject to the constraints $36x_1 + 6x_2 \geq 108$, $3x_1 + 12x_2 \geq 36$, $20x_1 + 10x_2 \geq 100$ and $x_1, x_2 \geq 0$

10.2 Network Analysis

Networks are diagrams easily visualized in transportation system like roads, railway lines, pipelines, blood vessels, etc.

A project will consist of a number of jobs and particular jobs can be started only after finishing some other jobs. There may be jobs which may not depend on some other jobs. Network scheduling is a technique which helps to determine the various sequences of jobs concerning a project and the project completion time. There are two basic planning and control techniques that use a network to complete a pre-determined schedule. They are Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM). The critical path method (CPM) was developed in 1957 by JE Kelly of Ramington R and M.R. Walker of Dupon to help schedule maintenance of chemical plants. CPM technique is generally applied to well known projects where the time schedule to perform the activities can exactly be determined.

Some important definitions in network

Activity:

An activity is a task or item of work to be done, that consumes time, effort, money or other resources. It lies between two events, called the starting event and ending event. An activity is represented by an arrow indicating the direction in which the events are to occur.

Event:

The beginning and end points of an activity are called events (or nodes). Event is a point in time and does not consume any resources. The beginning and completion

10_11th_BM-STAT_Ch-10-EM.indd 244 21-04-2020 12:32:53 PM